

Binary Search Definition

Binary Search is a searching algorithm used to find a specific element in a **sorted array**. It works by repeatedly dividing the array into **two halves** and comparing the target element with the middle element.

Algorithm

1. Set $low = 0$ and $high = n - 1$.
2. Find the middle element: $mid = (low + high) / 2$.
3. If $A[mid] == key$, the element is found.
4. If $key < A[mid]$, search the left half.
5. If $key > A[mid]$, search the right half.
6. Repeat until the element is found or $low > high$.

OUTPUT:

```
Enter number of elements: 5
Enter the elements in sorted order:
4
5
6
7
8
Enter the element to search: 7
Element found at position: 4
Execution Time: 1.5020370483398438e-05 seconds

Time Complexity:
Best Case : O(1)
Average Case: O(log n)
Worst Case : O(log n)
Space Complexity: O(1)

...Program finished with exit code 0
Press ENTER to exit console.
```

Time and space complexity

Case	Time Complexity	Space Complexity
Best case	$O(1)$	$O(1)$
Average case	$O(\log n)$	$O(1)$
Worst case	$O(\log n)$	$O(1)$